

Cass Staniec

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EDUCATION

Western Michigan University

Kalamazoo, MI

B.S. Candidate, Double Major: Aerospace Engineering & Mechanical Engineering. 2025 – Present

Relevant Coursework: Independent Research: authored and submitted a research paper titled ‘Alternate Propellants for Spacecraft Thrusters,’ examining design approaches for alternative forms of electrostatic propulsion systems.

Dakota High School

Macomb, MI

High School Diploma. GPA: 3.89.

June 2025

Relevant Coursework: Completed a four-year engineering program in three years, building hands-on skills in CAD, fabrication, welding, machining, and prototyping. MITES Competition: 1st Place, Go-Kart/Open Category (2025); 2nd Place, Hydraulic Wooden Fighting Structure/Open Category (2024); 7th Place, Guitar Player with Amplifier/Welding Sculpture (2025).

PROJECTS

Design Build Fly (DBF) — Glider Design & Construction

- Designed a glider in SolidWorks and translated the CAD model into a physical prototype using balsa wood and glue.
- Executed an end-to-end engineering workflow spanning CAD design, material selection, fabrication, and assembly.

3D-Printed I-Beam — Structural Design & Testing

- Designed an I-beam structure in SolidWorks and prepared it for additive manufacturing.
- 3D printed the component and conducted stress testing to evaluate structural performance.

Senior Engineering Go-Kart

- Contributed to a go-kart build through mechanical design, fabrication, assembly, and hands-on troubleshooting.
- Supported prototype construction and system assembly using physical components and shop processes.

SKILLS

Certifications: Certified SolidWorks Associate (CSWA)

CAD & Design: SolidWorks (3 yrs), AutoCAD (4 yrs), Autodesk Inventor (2 yrs), BendTech (2 yrs), 3D modeling, mechanical design, prototype development, design-to-fabrication workflows

Manufacturing & Fabrication: MIG and TIG welding, CNC mill/lathe operation and programming, precision machining, bending, cutting, drilling, mechanical assembly, fabrication-lab equipment

EXPERIENCE

Student Maintenance Mechanic

March 2026 – Present

Western Michigan University

- Maintain campus facilities and equipment using hand tools and machinery including welding, bending, cutting, and drilling equipment.
- Apply safe work procedures and attention to detail to uphold facility standards and support the university community.

Team Member

2024 – 2025

Target

- Maintained reliability and attention to detail in a fast-paced environment while balancing school and extracurricular engineering activities.

Ride Operations / Guest Services

2022 – 2024

Amusement Park

- Operated rides safely and developed familiarity with mechanical systems and troubleshooting procedures.
- Applied strict attention to operating and safety protocols in a high-responsibility environment.

INTERESTS

- Mechanical design, aerospace systems, alternative propulsion, welding and fabrication, automotive and motorsports engineering, prototyping